

Assignment-Chapter 5

1. Multiple Choices. (For each question, please select the most appropriate answer.)

1.1. Which of the following complexes may undergo associative substitution reactions?

- (a) $\text{V}(\text{CO})_6$ (b) $\text{Re}(\text{CO})_4(\text{NO})$ (c) $\text{CpRh}(\text{CO})_2$
 (d) two of them (e) all of them (f) none of them

e

1.2. Which of the following reactions would you expect to be dissociative substitution reactions?

- (a) $\text{Pt}(\text{CO})_4 + \text{PMe}_3 \longrightarrow \text{Pt}(\text{CO})_3(\text{PMe}_3) + \text{CO}$
 (b) $\text{HRh}(\text{CO})_4 + \text{PMe}_3 \longrightarrow \text{HRh}(\text{CO})_3(\text{PMe}_3) + \text{CO}$
 (c) $\text{CH}_3\text{Re}(\text{CO})_5 + \text{PMe}_3 \longrightarrow \text{CH}_3\text{Re}(\text{CO})_4(\text{PMe}_3) + \text{CO}$
 (d) $[\text{V}(\text{CO})_6]^- + \text{PMe}_3 \longrightarrow [\text{V}(\text{CO})_5(\text{PMe}_3)]^- + \text{CO}$
 (e) two of them (f) three of them (g) all of them (h) none of them

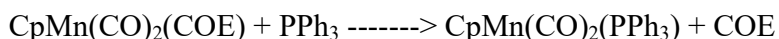
g

1.3. Which of the following dissociative substitution reactions would you expect to have the lowest reaction rate?

- (a) $\text{Ru}(\text{CO})_4(\text{PEt}_3) + \text{CO} \longrightarrow \text{Ru}(\text{CO})_5 + \text{PEt}_3$
 (b) $\text{Ru}(\text{CO})_4(\text{OEt}_2) + \text{CO} \longrightarrow \text{Ru}(\text{CO})_5 + \text{OEt}_2$
 (c) $\text{W}(\text{CO})_5(\text{PEt}_3) + \text{CO} \longrightarrow \text{W}(\text{CO})_6 + \text{PEt}_3$
 (d) $\text{W}(\text{CO})_5(\text{OEt}_2) + \text{CO} \longrightarrow \text{W}(\text{CO})_6 + \text{OEt}_2$

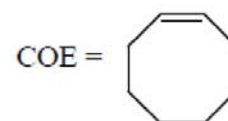
c

2. Propose a reaction mechanism for the following reaction



Your mechanism must be consistent with the following observations:

- (a) using excess PPh_3 , the rate is first order in Rh complex.
 (b) $\Delta S^\ddagger = -17.9 \text{ eu}$



Answer: Because $\Delta S^\ddagger < 0$, and the rate appears to be independent of $[\text{PPh}_3]$ using excess PPh_3 , the reaction is an associative substitution.

